Letter notes: CRN $\to$ generator, and how the numbers come out

Short reply to your 29 Apr 2026 note

1. The main claim, in one sentence

Given a chemical reaction network (CRN) / Doi–Peliti action, we write down a single ***generator*** *— a polynomial Dyson–Schwinger equation $G(z) = z\,\phi(G(z))$ — and read every diagrammatic number off it.* The rest of the framework is two tracks built on that one generator:

- **Track A (analytic combinatorics).** Counts, signs, symmetry factors, Z-factor pole rationals — by Lagrange inversion + bivariate marking + Hopf-antipode (Connes–Kreimer).
- **Track B (Connes–Kreimer / IBP).** The residual integrals after counting is factored out are reduced to a small *master basis*; their values are quoted from prior work or computed by HyperInt / tropical Monte Carlo.

The factorisation that ties the two tracks together is, for any 1PI amplitude,

$$\mathcal{A}_\Gamma \;=\; \underbrace{c_\Gamma}_{\text{count}} \cdot \underbrace{B_\Gamma}_{\text{bridge integral}} \cdot \underbrace{a_\Gamma}_{\text{algebra}},$$

with c_Γ and a_Γ rational outputs of Track A, and B_Γ the analytic residue handled by Track B.

2. Two closed forms: from CRN to diagrams, and from diagrams to numbers

There are *two* algorithmic generators in the framework, with different roles. Both are written down directly from the CRN.

2a. Liouvillian generator $\to$ amputated diagrams (enumeration)

This is the construction in Amarteifio (2019, PhD thesis) (Heisenberg–Weyl / Doi–Peliti). For a reaction network with stoichiometries $\sum_i k_{ri} A_i \to \sum_i l_{ri} A_i$ at rate k_r, the Doi shift produces, per reaction, the vertex generator

$$\boxed{Q_r \;=\; k_r \Big[\prod_i (1+\tilde\psi_i)^{l_{ri}} \;-\; \prod_i (1+\tilde\psi_i)^{k_{ri}}\Big] \prod_i \psi_i^{k_{ri}}.}$$

What it does on expansion. Expanding the binomials gives a finite *vertex dictionary*: each monomial $g_{\{m_i\},\{n_i\}} \prod_i \tilde\psi_i^{m_i} \psi_i^{n_i}$ is a vertex with m_i outgoing and n_i incoming legs of species A_i, with the rational coefficient g read off from the Doi binomials. The Feynman rules then mechanically connect ψ-legs to $\tilde\psi$-legs through propagators, producing all amputated 1PI diagrams. **This is the part that algorithmically (or by hand) enumerates the diagrams** — it is the standard Doi–Peliti expansion, written in a form that is mechanical rather than handcrafted per system.

For Reggeon DP, $A \to 2A$ at rate λg and $2A \to A$ at rate λg give, after the Doi shift,

$$Q \;=\; -\lambda g\,\tilde\psi(1+\tilde\psi)\psi \;+\; \lambda g\,\tilde\psi(1+\tilde\psi)\psi^2 \;=\; \lambda g\big(\tilde\psi\psi^2 - \tilde\psi^2\psi\big) \;+\; \text{(free part)},$$

recovering the JT05 cubic vertices $\frac{\lambda g}{2}(\tilde{\psi}\psi^2 - \tilde{\psi}^2\psi)$ with their relative sign.

What this generator does *not* fix automatically. The amputated graphs are enumerated, but the per-graph *symmetry factors* come from explicit graph-automorphism counting (or from Wick contractions): the generator gives you the diagrams; you still have to figure out $|\mathrm{Aut}(\Gamma)|^{-1}$. That is the residual hand-step in this layer.

2b. Lagrange / DSE generator → counts and symmetry factors (CFAC)

The CFAC closure adds a second generator that handles *both* the count *and* the symmetry factors in one move. For Reggeon DP the Liouvillian above gives the polynomial DSE

$$\boxed{G(z) \;=\; z\,\phi(G(z)), \qquad \phi(G) \;=\; 1 + G^2.}$$

Reading the diagram counts off the generator (and the symmetry factors). Lagrange inversion (Flajolet–Sedgewick A.6),

$$[z^n]\,G(z) \;=\; \tfrac{1}{n}\,[G^{n-1}]\,\phi(G)^n,$$

gives, for $\phi(G) = 1 + G^2$:

$$[z^3]G = \tfrac{1}{3}\binom{3}{1} = 1, \qquad [z^5]G = \tfrac{1}{5}\binom{5}{2} = 2, \qquad [z^7]G = \tfrac{1}{7}\binom{7}{3} = 5.$$

That is the *entire* 1- and 2-loop diagram inventory: $1 + 2 + 5 = 8$ topologies, no graphs drawn, **symmetry factors automatic** (they appear as the Lagrange-inversion overcount, 1/2! on bubble, 1/3! on sunset, etc.). *This is the gain over (2a)*: the Liouvillian gets you to amputated diagrams; the Lagrange closure removes the automorphism bookkeeping as well.

Reading the rapidity signs off the generator. Bivariate marking $G(z, \alpha) = z(1 + \alpha G^2)$ at $\alpha = -1$ gives the signed Reggeon count; $[z^n\alpha^k]G$ counts trees with k opposite-sign vertex pairs.

Reading the double poles off the generator (Hopf antipode). The Connes–Kreimer antipode evaluation closes *every* 2-loop double pole in closed form:

$$Z_X^{(2,2)} \;=\; \tfrac{1}{2}\,a_X^{(1)}\big(\beta_1 + a_X^{(1)}\big), \qquad \beta_1 = a_u^{(1)} - 2\,a_\psi^{(1)} = \tfrac{3}{2},$$

with $a^{(1)} = \{1/4, 1/8, 1/2, 2\}$ from the 1-loop algebra. Result: $\{Z_\psi, Z_\lambda, Z_\tau, Z_u\}^{(2,2)} = \{7/32,\ 13/128,\ 1/2,\ 7/2\}$, matching JT05 Eq. (57) exactly.

Reading the exponents off the generator. Pipe through the Täuber relation, $\gamma_X(u) = -u\,\partial_u(\text{simple-pole residue of } \ln Z_X)$:

- $\gamma(u), \zeta(u), \kappa(u)$ at 2 loops — match JT05 Eq. (58a/b/c) with *zero residual* (`eq58_from_z.py`).
- $\beta(u)$ at 1-loop is $3u^2/2$ from β_1 above; the 2-loop coefficient is bookkeeping (Vasil'ev MSbar pole-cancellation), not yet written symbolically.
- Fixed point u^* and exponents $\{\eta, z, \nu, \beta_{\mathsf{DP}}\}$ at JT05 Eq. (60) — *zero residual* (`rdft/ac/gribov/two_loop.p`

The only analytic input that is *not* derived from ϕ is the value of three master integrals $\{B_2, B_3^{\mathsf{sun}}, B_V\}$. CFAC reduces ~ 10 distinct loop integrals to those three; computing their numerical values is grunt work where Track B (HyperInt / tropical MC / hand) takes over.

3. Where QFT enters: relevance filters

QFT supplies the filtering rules:

- which external-leg sector is being renormalised (2-point for Z_ψ, 3-point for Z_u),
- which loop order,
- the 1PI restriction (no reducible trees),
- power-counting for UV relevance ($\omega = dL - 2I + n_\partial \geq 0$).

In our current pipeline these are *hand-mapped* on top of the Lagrange counts (e.g. $[z^7] = 5 = 3$ 1PI vertex topologies $+\,2$ reducible trees, in `gribov.tex` §3.3).

Hypothetically we can fold QFT into the generator itself. The proposal — and we are testing it on a separate track, not yet adopted — is to enrich the generator with marker variables:

$$\mathcal{G}(z;\, \xi_\psi, \xi_{\tilde\psi};\, \alpha) \;=\; \sum c_{n,e_\psi,e_{\tilde\psi},s}\, z^n\, \xi_\psi^{e_\psi}\, \xi_{\tilde\psi}^{e_{\tilde\psi}}\, \alpha^s,$$

take its Legendre transform $\mathcal{W} = \ln \mathcal{Z} \to \Gamma$ to project onto 1PI, and apply a power-counting inequality on monomial degrees as a filter. Each Z-factor would then be a specific bidegree projection of Γ_{1PI}, with no human-mapped topology table.

POC, and the honest verdict. The Legendre track is implemented for 0-d Reggeon DP in `poc_legendre.py` (this letter directory; see also the Legendre tutorial in `docs/`). It runs end-to-end and reproduces the 3+2 split (3 1PI vertex graphs + 2 reducibles) at g^5 that gribov.tex hand-mapped. **But**: for a 2-loop calculation with ~ 5 topologies, hand-mapping is faster than building the autonomous selector. The autonomous filter pays off at higher loop orders or for theories without a curated topology table (your H-vertex model, Manna, BARW). *So: yes the filter can be combinatorial; no, it doesn't always save time at low orders.*

4. Q1: your 2^n ladder, in our framework

You wrote (paraphrasing): the four-term coupling

$$\left[\ \overset{m_1 n_1}{\underset{m_2 n_2}{\Join}}\ \right] = \delta_{m_2,n_2}\delta_{m_1-1,n_1+1} + \delta_{m_2,n_2}\, n_1\, \delta_{m_1-1,n_1-1}$$
$$- \delta_{m_2,n_2+1}\delta_{m_1-1,n_1+1} - \delta_{m_2,n_2-1}\, n_2\, \delta_{m_1-1,n_1}$$

gives an n-loop chain

$$\sum_{\ell_1,\dots,\ell_{2n}} \overset{1\ \ell_1}{\underset{0\ \ell_2}{\Join}} \cdot \left(\overset{\ell_1\ \ell_3}{\underset{\ell_2\ \ell_4}{\Join}} + \overset{\ell_2\ \ell_4}{\underset{\ell_1\ \ell_3}{\Join}} \right) \cdots$$
$$\cdots \left(\overset{\ell_{2n-1} 1}{\underset{\ell_{2n}\ 0}{\Join}} + \overset{\ell_{2n}\ 0}{\underset{\ell_{2n-1} 1}{\Join}} \right) = 2^n,$$

verified by Mathematica through several orders, with the question: *can analytic combinatorics see this 2^n structurally? What is $T(z)$, and how is it calculated? When I try this I get an infinite matrix or a generating function in four variables, so it's a mess.*

Answer in one box. Encode each leg occupation by a marker — z_i for input rail i, w_j for output rail j — and let $\widehat{V}(z_1, z_2; w_1, w_2) = \sum V(m_1, m_2; n_1, n_2) z_1^{m_1} z_2^{m_2} w_1^{n_1} w_2^{n_2}$. The four-variable polynomial *is* unbounded as you said, but the boundary $|(1,0)\rangle$ projects onto bidegree $(1,1)$, on which only T2 and T4 survive:

$$\widehat{V}\big|_{(1,1)}(z, w) \;=\; z_1 w_1 - z_1 w_2.$$

Add the rail-flip $\widetilde{V}$ ($z_1 \leftrightarrow z_2,\ w_1 \leftrightarrow w_2$):

$$\boxed{\widehat{T}(z,w) \;=\; \big(\widehat{V} + \widetilde{V}\big)\big|_{(1,1)} \;=\; z_1 w_1 - z_1 w_2 - z_2 w_1 + z_2 w_2 \;=\; (z_1 - z_2)(w_1 - w_2).}$$

Read off the eigenvalue. The factorisation is rank-1 *by inspection*: input mode $z_1 - z_2$ couples to output mode $w_1 - w_2$ and to nothing else. Substituting the antisymmetric input $z_1 = 1, z_2 = -1$ gives $\widehat{T} = 2(w_1 - w_2)$ — **eigenvalue $\lambda = 2$ on the antisymmetric mode.** Symmetric input $z_1 = z_2 = 1$ gives 0 (kernel). So

$$\big\langle \text{boundary}_L \big| \widehat{T}^{\,n} \big| \text{boundary}_R \big\rangle \;=\; \lambda^n \;=\; \boxed{2^n.}$$

Three structural inputs predict $\{0, 2\}$ before any matrix is written.

1. *Boundary* $\Rightarrow$ *sector dim* $= 2$. The boundary fixes total particle number $N = 1$; $\#\{(m_1, m_2) \in \mathbb{Z}_{\geq 0}^2 : m_1 + m_2 = 1\} = 2$.
2. *T2 is the rail-1 number operator* $\Rightarrow$ $\mathrm{Tr}(T) = 2$. Term 2 is the only diagonal-preserving term, with weight $n_1 = m_1$. Together with rail-flip $\widetilde{V}$'s diagonal weight m_2: $T_{\text{diag}}(m_1, m_2) = m_1 + m_2 = N$. So $\mathrm{Tr}(T) = N \cdot \dim = 1 \cdot 2 = 2$.
3. *Rank-*1 *image* $\Rightarrow$ $\det(T) = 0$. On the single-particle sector, T2/T4 require $m_1 \geq 1$, so V annihilates $|(0,1)\rangle$.

A $2{\times}2$ matrix with $\mathrm{Tr} = 2$ and $\det = 0$ has eigenvalues $\{0, 2\}$. Both numbers came from structure, not arithmetic.

Direct answers to your three sub-questions.

- *Why* $2{\times}2$*, not infinite?* The boundary $|(1,0)\rangle$ at both ends restricts to the bidegree-$(1,1)$ subspace; that has dimension $2{\times}2 = 4$, which the rail-flip $\mathbb{Z}_2$ symmetry collapses to a rank-1 bilinear form.
- *What is $T(z)$?* It is the bidegree-$(1,1)$ projection of $\widehat{V} + \widetilde{V}$, with no z-dependence beyond $\{z_1, z_2, w_1, w_2\}$ used as basis labels. Concretely, $\widehat{T}(z, w) = (z_1 - z_2)(w_1 - w_2)$.
- *How is it calculated?* Walk the four Kronecker conditions of V at bidegree $(1,1)$ (a paragraph), add the rail-flip, factor. Half a page, no Mathematica.

Verification. The script `verify_2x2.py` (this directory) runs both the polynomial collapse symbolically (sympy: $\widehat{T} = (z_1 - z_2)(w_1 - w_2)$, eigenvalue 2) and the brute-force occupation sum: $2, 4, 8, 16, 32$ for $n = 1, \ldots, 5$, matching your Mathematica.

5. The connection to your φ^4 point

The $4 \cdot 3$ and $\binom{4}{2}^2 \cdot 2 \cdot 4!/2$ Wick weights are the bidegree-$(2,2)$ and $(4,4)$ coefficient extractions of the source-coupled $\mathcal{Z}(g, r; J)$. The 2^n ladder is the bidegree-$(1,1)$ eigenvalue iteration of your $\widehat{V}$. The Gribov tree counts $\{1, 2, 5\}$ are $[z^n]G$ of $G = z(1 + G^2)$. *Two observables, three theories, one* *machine*: vertex polynomial → boundary-projected sector → closed-form residue / eigenvalue. Counting on one end, exponents on the other, with the bridge integrals as the only analytic residue in between.

Companion notes Q1 and Q2 in this directory go through the 2^n *derivation (Q1) and the JT05 Eq. (57)–(60) reproduction (Q2) in fuller detail.*